DPU vs NIC-mode (BF3-Bond) Performance Comparison

Complete Analysis - All Test Types and Topologies (with CPU Consumption)

Executive Summary

Overall Results: 39 matching test configurations compared between DPU and NIC-mode (BF3-Bond)
DPU wins 28/39 tests (71.8%) | NIC-mode wins 11/39 tests (28.2%)

MATCHING CONFIGURATIONS

39/42

92.9% coverage

DPU WINS

28/39

71.8% win rate

NIC-MODE WINS

11/39

28.2% win rate

⚡ CPU Efficiency - Internode Traffic

DPU uses 34.5% fewer CPUs than NIC-mode for internode workloads

- DPU average: **39.0 busy CPUs** @ 272.41 Gbps = **6.99 Gbps/CPU**
- NIC-mode average: **59.5 busy CPUs** @ 233.58 Gbps = **3.93 Gbps/CPU**
- **DPU is 78% more CPU-efficient** (delivers 1.78x more throughput per CPU)

Key Findings

- DPU dominates **internode** workloads with superior throughput AND CPU efficiency
- NIC-mode excels at **intranode** workloads (same-host communication)
- Internode TCP Stream: DPU wins 94.4% with +16.6% throughput and -34.5% CPU usage
- Intranode TCP Stream: NIC-mode wins 100% with +74% throughput
- DPU offloads network processing to hardware, freeing host CPUs

Performance by Workload Type

Internode TCP Stream (18 tests)

DPU WIN RATE

94.4%

17 out of 18 tests

DPU AVERAGE

272.41 Gbps

Throughput

NIC-MODE AVERAGE

233.58 Gbps

Throughput

DPU ADVANTAGE

+16.6%

+38.83 Gbps average

DPU AVG CPU

39.0

Busy CPUs

NIC-MODE AVG CPU

59.5

Busy CPUs

DPU CPU EFFICIENCY

6.99

Gbps per CPU

NIC-MODE CPU EFFICIENCY

3.93

Gbps per CPU

Best DPU performance: 329.80 Gbps using only 44.8 busy CPUs (7.36 Gbps/CPU)
Configuration: 2 guaranteed CPUs, 134KB packets, 32 threads
vs NIC-mode: 211.87 Gbps using 60.9 busy CPUs (3.48 Gbps/CPU)
Advantage: +55.7% throughput with -26% fewer busy CPUs

Intranode TCP Stream (4 tests)

NIC-MODE WIN RATE

100%

4 out of 4 tests

DPU AVERAGE

285.72 Gbps

Throughput

NIC-MODE AVERAGE

497.03 Gbps

Throughput

DPU DEFICIT

-42.5%

-211.32 Gbps average

DPU AVG CPU

33.6

Busy CPUs

NIC-MODE AVG CPU

59.6

Busy CPUs

DPU CPU EFFICIENCY

8.51

Gbps per CPU

NIC-MODE CPU EFFICIENCY

8.34

Gbps per CPU

Intranode observation: NIC-mode achieves +74% higher throughput despite using +77% more CPUs. For same-host traffic, kernel optimizations outweigh DPU hardware offload benefits.

Internode TCP/UDP Request-Response (12 tests)

DPU WIN RATE

75.0%

9 out of 12 tests

DPU AVERAGE

765,284

transactions/sec

NIC-MODE AVERAGE

499,044

transactions/sec

DPU ADVANTAGE

+53.4%

+266K trans/sec

DPU AVG CPU

25.6

Busy CPUs

NIC-MODE AVG CPU

27.9

Busy CPUs

DPU CPU EFFICIENCY

29,889

trans per CPU

NIC-MODE CPU EFFICIENCY

17,893

trans per CPU

Intranode TCP/UDP Request-Response (5 tests)

NIC-MODE WIN RATE

60.0%

3 out of 5 tests

DPU AVERAGE

1,527,300

transactions/sec

NIC-MODE AVERAGE

2,855,110

transactions/sec

DPU DEFICIT

-46.5%

-1.3M trans/sec

DPU AVG CPU

41.9

Busy CPUs

NIC-MODE AVG CPU

41.8

Busy CPUs

DPU CPU EFFICIENCY

36,442

trans per CPU

NIC-MODE CPU EFFICIENCY

68,315

trans per CPU

Detailed Comparison - Internode TCP Stream (with CPU Data)

Threads	Wsize	CPU	Perf Profile	DPU Gbps	DPU Busy CPU	DPU Gbps/CPU	NIC-mode Gbps	NIC-mode Busy CPU	NIC-mode Gbps/CPU	Throughput Difference	CPU Difference
1	32KB	2	single-numa	236.59	19.7	**11.98**	249.01	34.8	7.15	**-5.0%**	**-15.1**
1	32KB	2(Gu)	single-numa	217.41	20.0	**10.88**	208.03	34.4	6.05	**+4.5%**	**-14.4**
1	134KB	2(Gu)	single-numa	222.17	19.3	**11.50**	211.35	34.6	6.10	**+5.1%**	**-15.3**
32	512B	2	single-numa	301.67	86.9	3.47	223.60	96.3	2.32	**+34.9%**	**-9.4**
32	512B	29	single-numa	236.03	63.0	3.75	211.58	74.7	2.83	**+11.6%**	**-11.7**
32	512B	29(Gu)	single-numa	235.06	52.1	**4.51**	198.55	70.9	2.80	**+18.4%**	**-18.8**
32	512B	58	None	251.31	62.6	**4.02**	213.78	74.1	2.88	**+17.6%**	**-11.6**
32	512B	58(Gu)	None	234.23	58.9	**3.98**	214.89	72.5	2.96	**+9.0%**	**-13.6**
32	32KB	29	single-numa	303.15	33.7	**9.00**	250.09	56.6	4.42	**+21.2%**	**-22.9**
32	32KB	29(Gu)	single-numa	260.90	26.8	**9.75**	232.42	56.9	4.09	**+12.3%**	**-30.1**
32	32KB	58	None	314.08	26.0	**12.08**	247.93	59.3	4.18	**+26.7%**	**-33.3**
32	32KB	58(Gu)	None	311.15	22.0	**14.16**	265.82	53.3	4.98	**+17.1%**	**-31.4**
32	134KB	2	single-numa	270.25	64.0	4.22	233.59	80.1	2.92	**+15.7%**	**-16.1**
32	134KB	2(Gu)	single-numa	**329.80**	**44.8**	**7.36**	211.87	60.9	3.48	**+55.7%**	**-16.1**
32	134KB	29	single-numa	296.60	30.9	**9.58**	255.00	53.3	4.78	**+16.3%**	**-22.3**
32	134KB	29(Gu)	single-numa	256.12	25.6	**10.00**	236.43	57.5	4.11	**+8.3%**	**-31.9**
32	134KB	58	None	312.74	24.0	**13.04**	248.88	57.4	4.34	**+25.7%**	**-33.4**
32	134KB	58(Gu)	None	314.17	21.3	**14.78**	291.61	42.6	6.84	**+7.7%**	**-21.4**

Key CPU Efficiency Observations

- DPU saves **9 to 33 busy CPUs** per test compared to NIC-mode
- DPU achieves **3.5x to 3.4x better Gbps/CPU** in most configurations

- Best DPU efficiency: **14.78 Gbps/CPU** (58 Gu CPUs, 134KB) vs NIC-mode 6.84 Gbps/CPU
- Even in the one test where NIC-mode wins throughput, DPU still uses 43% fewer CPUs

Detailed Comparison - Intranode TCP Stream (with CPU Data)

Intranode Stream CPU Efficiency: DPU saves 26.0 CPUs (43.7%) but NIC-mode achieves +74% higher throughput. For same-host traffic, NIC-mode's kernel optimizations deliver better performance despite higher CPU usage.

Protocol	Type	Threads	Wsize	CPU	Perf Profile	DPU Gbps	DPU Busy CPU	DPU Gbps/CPU	NIC-mode Gbps	NIC-mode Busy CPU	NIC-mode Gbps/CPU	Throughput Diff	CPU Diff
TCP	stream	32	32KB	2	single-numa-node	254.74	54.7	4.66	480.21	68.9	**6.97**	**-47.0%**	**-14.1**
TCP	stream	32	32KB	28(Gu)	None	307.78	28.9	**10.63**	452.33	58.9	7.69	**-32.0%**	**-29.9**
TCP	stream	32	32KB	29	None	276.87	26.1	**10.61**	379.35	55.9	6.78	**-27.0%**	**-29.8**
TCP	stream	32	134KB	28(Gu)	None	303.49	24.6	**12.34**	676.25	54.8	12.33	**-55.1%**	**-30.2**

Key Intranode CPU Observations

- DPU is more CPU-efficient per Gbps in 3 out of 4 tests
- DPU saves 14-30 busy CPUs per workload
- NIC-mode achieves higher absolute throughput despite higher CPU usage
- For same-host traffic, kernel fast path optimizations outweigh DPU hardware offload

Detailed Comparison - Internode Request-Response (with CPU Data)

Internode RR CPU Efficiency: DPU saves only 2.3 CPUs (8.2%) on average. Mixed CPU results - DPU uses fewer CPUs in high-performance scenarios but more CPUs in some low-throughput tests.

Protocol	Type	Threads	Wsize	CPU	Perf	DPU trans/sec	DPU Busy CPU	DPU trans/CPU	NIC-mode trans/sec	NIC-mode Busy CPU	NIC-mode trans/CPU	Throughput Diff	CPU Diff
TCP	rr	1	1KB	2(Gu)	single-numa	693,517	26.7	**25,971**	211,986	17.7	11,946	**+227.2%**	+9.0
TCP	rr	32	1KB	2(Gu)	single-numa	1,297,879	52.0	**24,963**	1,509,193	92.0	16,399	**-14.0%**	**-40.0**
TCP	rr	32	1KB	29	single-numa	1,500,020	64.3	**23,341**	538,925	32.8	16,413	**+178.3%**	+31.4
TCP	rr	32	1KB	29(Gu)	single-numa	1,512,752	57.5	**26,289**	492,978	34.6	14,240	**+206.9%**	+22.9
TCP	rr	32	1KB	58	None	676,050	17.5	**38,703**	541,109	33.2	16,306	**+24.9%**	**-15.7**
TCP	rr	32	1KB	58(Gu)	None	719,791	16.7	**42,991**	567,267	32.1	17,686	**+26.9%**	**-15.3**
UDP	crr	1	512B	2	single-numa	3,248	1.3	**2,574**	5,764	6.6	867	**-43.6%**	**-5.4**
UDP	rr	1	1KB	2	single-numa	334,988	14.6	**22,975**	83,914	9.0	9,361	**+299.2%**	+5.6
UDP	rr	1	1KB	2(Gu)	single-numa	264,854	10.5	**25,252**	79,809	9.5	8,396	**+231.9%**	+1.0
UDP	rr	32	1KB	2(Gu)	single-numa	705,453	7.4	**95,916**	853,087	27.8	30,673	**-17.3%**	**-20.5**
UDP	rr	32	1KB	29	single-numa	160,122	10.2	**15,740**	87,583	7.5	11,646	**+82.8%**	+2.7
UDP	rr	32	1KB	4	None	1,314,735	28.7	**45,843**	1,016,910	31.8	32,026	**+29.3%**	**-3.1**

Key Internode RR CPU Observations

- DPU achieves better trans/CPU efficiency in 9 out of 12 tests
- DPU saves up to 40 busy CPUs in high-load scenarios (32T, 2Gu CPUs)

- DPU wins 75% of tests with +53.4% average throughput
- Best DPU efficiency: 95,916 trans/CPU (UDP RR, 32T, 2Gu CPUs)

Detailed Comparison - Intranode Request-Response (with CPU Data)

Intranode RR CPU Efficiency: Virtually no CPU difference (-0.1 CPUs). Mixed results with NIC-mode winning 60% of tests through superior intranode optimizations.

Protocol	Type	Threads	Wsize	CPU	Perf	DPU trans/sec	DPU Busy CPU	DPU trans/CPU	NIC-mode trans/sec	NIC-mode Busy CPU	NIC-mode trans/CPU	Throughput Diff	CPU Diff
TCP	rr	32	1KB	14	single-numa	1,307,851	47.0	27,803	1,329,297	38.0	**35,004**	**-1.6%**	+9.1
TCP	rr	32	1KB	2	single-numa	1,119,539	48.5	23,091	5,647,312	69.0	**81,897**	**-80.2%**	**-20.5**
TCP	rr	32	1KB	4(Gu)	None	1,230,638	31.7	38,761	5,718,041	66.6	**85,888**	**-78.5%**	**-34.8**
UDP	rr	32	1KB	2	single-numa	1,659,715	62.5	26,535	799,946	21.4	**37,353**	**+107.5%**	+41.1
UDP	rr	32	1KB	2(Gu)	single-numa	2,318,759	19.7	**117,522**	780,955	14.0	55,607	**+196.9%**	+5.7

Key Intranode RR CPU Observations

- NIC-mode achieves exceptional trans/CPU efficiency for TCP RR (up to 85,888 trans/CPU)
- DPU wins both UDP RR tests with 2x-3x better throughput
- For intranode TCP RR, NIC-mode's kernel fast path delivers 4-5x higher throughput
- Mixed CPU usage patterns - no clear winner in CPU efficiency

Key Insights

1. Topology Makes the Difference

- **Internode (cross-host):** DPU dominates with 86.7% win rate across all test types
- **Intranode (same-host):** NIC-mode dominates with 77.8% win rate
- DPU hardware offload excels for network traffic but adds overhead for local traffic

2. DPU's CPU Efficiency Advantage

- **34.5% fewer busy CPUs** for internode workloads (39 vs 59.5 average)
- **78% better CPU efficiency** - delivers 6.99 Gbps/CPU vs NIC-mode's 3.93 Gbps/CPU
- Hardware offload frees host CPUs for application workloads
- Saves 9-33 busy CPUs per workload, critical in CPU-constrained environments
- Superior CPU efficiency even when throughput is similar to NIC-mode

3. DPU Advantages for Internode Traffic

- Consistent performance gains across all packet sizes (512B to 134KB)
- Strongest advantage with medium CPU allocations (29 CPUs: saves 23-30 busy CPUs)
- Best result: 329.80 Gbps with just 2 guaranteed CPUs using 44.8 busy CPUs
- Superior performance for both stream and request-response workloads

4. NIC-mode Advantages for Intranode Traffic

- 74% faster for intranode stream workloads (497 vs 286 Gbps average)
- 87% faster for intranode request-response (2.9M vs 1.5M trans/sec)
- Benefits from kernel-level optimizations for same-host communication
- No DPU overhead for local pod-to-pod traffic

Recommendations

Choose DPU for:

- **Internode workloads** - Cross-host network traffic (+16.6% throughput, -34.5% CPU)
- **CPU-constrained environments** - Save 20-30 busy CPUs per workload
- **Multi-threaded applications** - 32+ threads performing network I/O
- **Maximum CPU efficiency** - Up to 14.78 Gbps/CPU vs NIC-mode's 6.84 Gbps/CPU
- **Small to large packet sizes** - Consistent gains across 512B-134KB
- **Request-response patterns** - 53% advantage for internode RR workloads
- **Cost optimization** - Lower CPU consumption reduces infrastructure costs

Choose NIC-mode (BF3-Bond) for:

- **Intranode workloads** - Same-host pod-to-pod communication (+74% throughput)
- **Mixed workloads** - Combination of local and network traffic
- **Simplicity** - No DPU hardware or configuration required
- **High intranode throughput** - Up to 676 Gbps for local traffic
- **CPU abundance** - When CPU resources are not a constraint

⚠ Note: Unmatched Test Configurations

3 out of 42 NIC-mode tests (7.1%) do not have matching DPU equivalents:

- **Internode TCP RR** - 32 threads, 1024B, CPU=2, perf=None (NIC-mode: 1.82M trans/sec)
- **Internode TCP Stream** - 1 thread, 32KB, CPU=2, perf=None (NIC-mode: 258.60 Gbps)
- **Internode TCP Stream** - 1 thread, 134KB, CPU=2, perf=None (NIC-mode: 272.95 Gbps)

Reason: All 3 tests are from the NO-PAO group with CPU=2 and perf=None. The DPU batch tested NO-PAO configurations with CPU=4 instead of CPU=2. Two of these tests have DPU equivalents with perf=single-numa-node instead of perf=None.
Source: regulus_data path = `1_GROUP/NO-PAO/4IP/INTER-NODE/TCP/N-POD/run-thu-2026-03-12-10:25:09/`

Summary Statistics

Metric	Value
Total BF3-Bond Tests	42 unique configurations
Matching Configurations	39/42 (92.9%)
Overall DPU Win Rate	28/39 (71.8%)
Internode DPU Win Rate	26/30 (86.7%)
Intranode NIC-mode Win Rate	7/9 (77.8%)
DPU Peak Throughput	329.80 Gbps (internode, 44.8 busy CPUs)
NIC-mode Peak Throughput	676.25 Gbps (intranode, N/A busy CPUs)
DPU Avg Busy CPUs (Internode)	39.0 CPUs
NIC-mode Avg Busy CPUs (Internode)	59.5 CPUs
DPU CPU Efficiency (Internode)	6.99 Gbps/CPU (78% better than NIC-mode)
NIC-mode CPU Efficiency (Internode)	3.93 Gbps/CPU

Generated on March 16, 2026
Data source: OpenSearch regulus-results-* indices
Batches compared: DPU (c4fc25af) vs NIC-mode BF3-Bond (2366a006)
Report location: /home/hnhan/NVD-DPU/bond-nic-regulus/REPORT/